

Origin of Quantum Physics

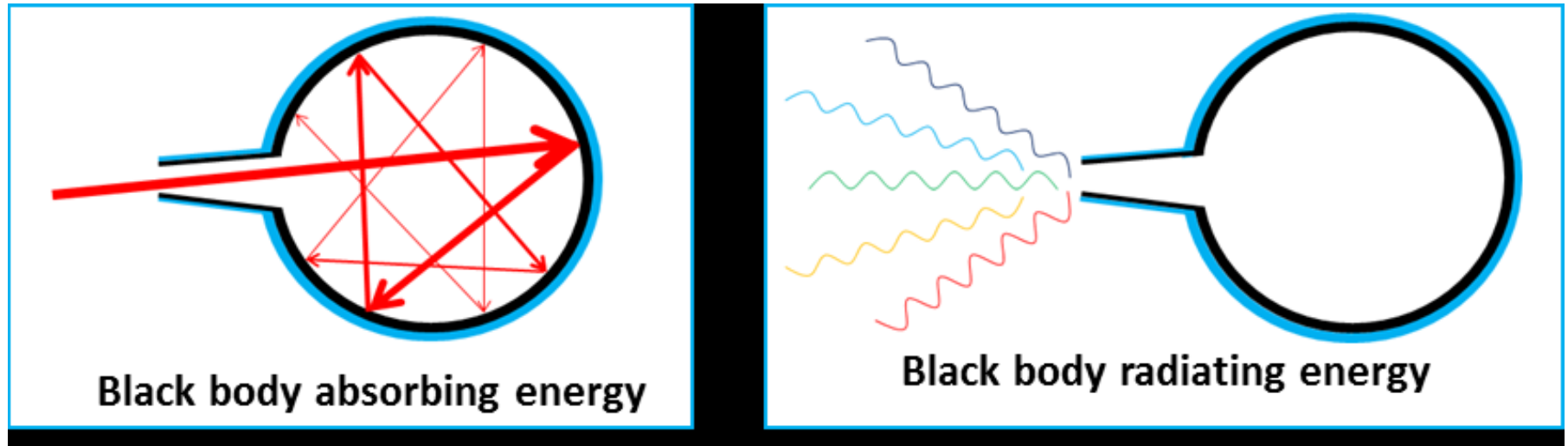
UNIT-1

•Lecture -2

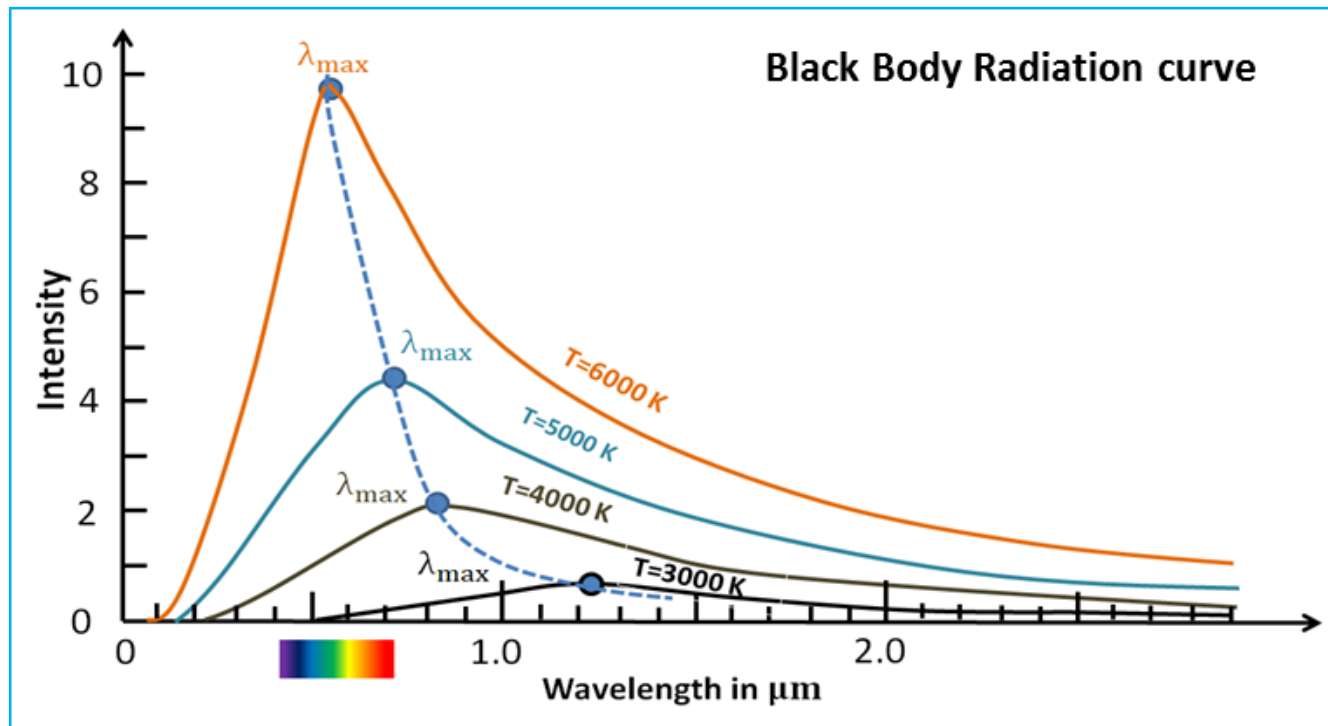
Blackbody radiation

Rayleigh scattering

Origin of Quantum Physics



The concept of quantum physics was first used by Max Planck to explain the result obtained black body radiation experiment. A black body is designed to absorb 99.9% of radiation incident on it. This can be done with a hollow sphere with a narrow hole. Inner surface of this hollow sphere is coated with carbon black. When radiation enters through a narrow hole, it hits the wall of the sphere and get almost fully absorbed. Remaining radiation is reflected back. And collide again with the wall. After number of reflections, almost all the radiation is absorbed. Due to absorption of radiation, temperature of the body increases and it starts emitting radiation from narrow hole. This radiation contains all wavelengths. It is called as black body radiation.



When intensity vs wavelength curve is plotted, it is found that radiation for longer wavelengths is very low. For shorter wavelengths, intensity is more. For a particular wavelength, λ_{max} , maximum energy is radiated. This is called peak wavelength.

Even though all wavelengths are emitted, there is a maximum intensity only for a peak wavelength. Black body appears of the color corresponding to that wavelength. - $\lambda_{\text{max}} \propto \frac{1}{T}$

If we increase temperature of the black body, λ_{max} shifts towards left side and the colour of the black body changes. This shift in the peak wavelength was explained by Wien's displacement law

Blackbody radiation

A black body is an idealized object that absorbs all electromagnetic radiation it comes in contact with.

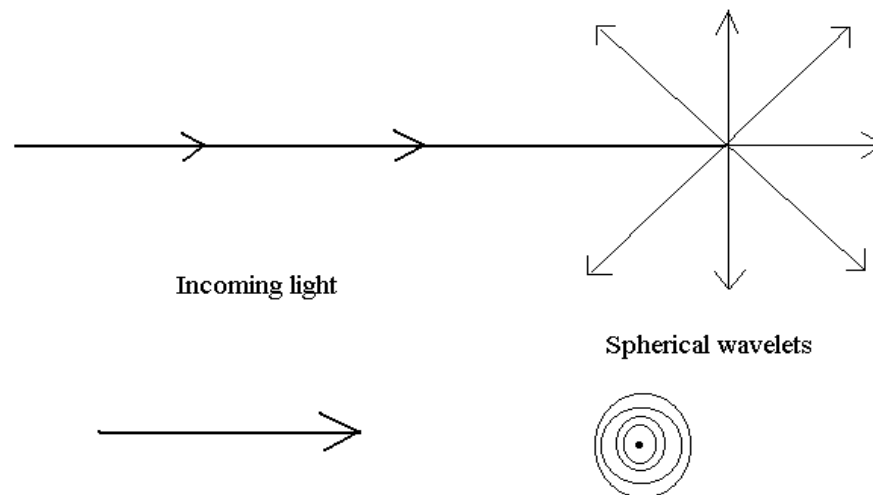
- ✓ Energy radiated by any object or system that absorbs all incident radiation.
- ✓ The amount of radiation of a particular wavelength given off from a surface include the type of material absorbing the incident radiation and the temperature of the surface.
- ✓ Different materials can give off different amounts of energy, even if the materials are at the same temperature.
- ✓ An object that gives off the maximum amount of energy for its absolute temperature is termed a blackbody

A *blackbody* allows *all* incident radiation to pass into it (no reflected energy) and internally absorbs *all* the incident radiation (no energy transmitted through the body). This is true for radiation of all wavelengths and for all angles of incidence. Hence the blackbody is a *perfect absorber for all incident radiation*

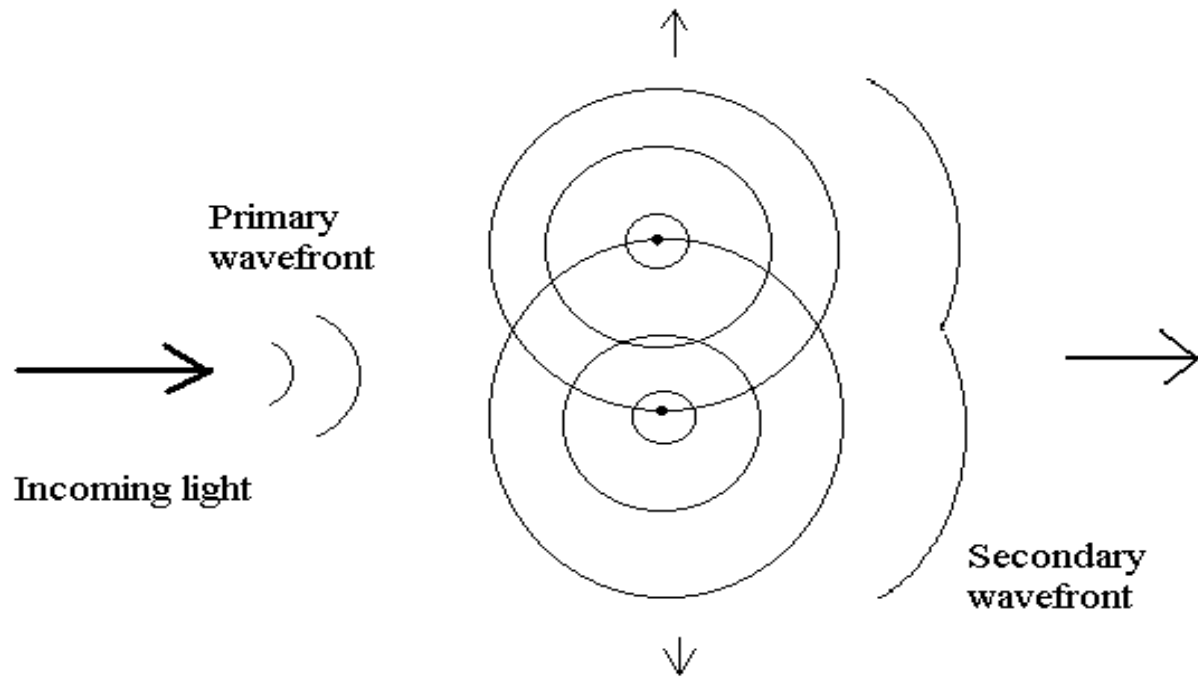
Rayleigh scattering

Rayleigh scattering of light by particles that are less than $1/15$ of the wavelength of the light. Atoms and ordinary molecules which have a diameter of less than one nanometer are included in Rayleigh scattering

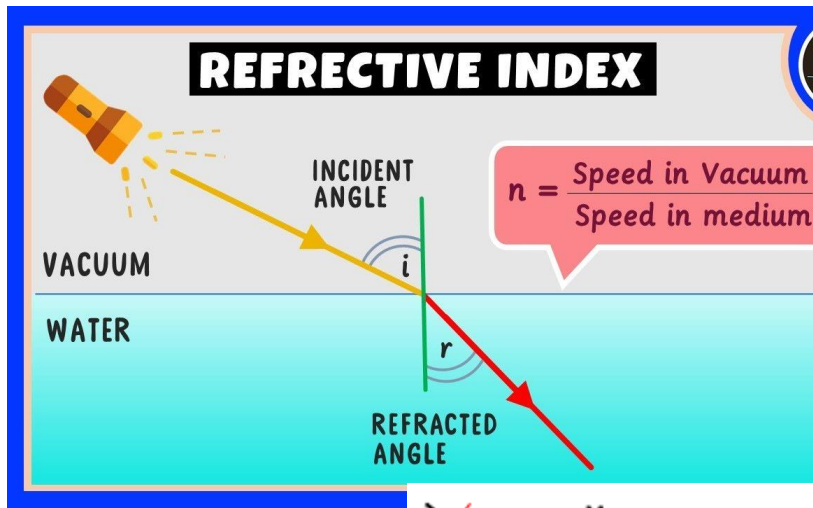
- When light travels through a medium, it interacts with the medium and this causes the scattering of light.
- Photons are absorbed by molecules in the medium, and this causes the molecules to vibrate and then re-emit the photons. When the photons are re-emitted, they have the same frequency and wavelength, but are re-emitted in random directions.
- There are a large number of photons interacting with the molecules, and so photons are emitted from molecules in all directions and appear to be spherical wavelets of light



- The scattering of light increases with frequency. Therefore light with higher frequencies undergoes more lateral scattering than light with lower frequencies.
- In media with random, widely spaced molecules, the light is scattered randomly in all directions except for the direction that the incident light was traveling. Therefore, lateral scattering is possible and observable
 - In dense, uniform media, scattered light will interfere constructively in the forward direction, but interfere destructively in all other directions. Therefore, little or no lateral scattering will be observed.



Index of Refraction (n): When light is transmitted through a material, each photon travels at the speed of light, c . However, Rayleigh scattering causes the transmitted wave to be out of phase with the free-space (initial) wave. The light is scattered many times as it travels through the medium. Each time the light is scattered, the scattering causes a phase shift. This translates to a change in the phase velocity (v) within the medium, of the beam from c . Therefore, there is an index of refraction for homogeneous materials, $n = c/v$.

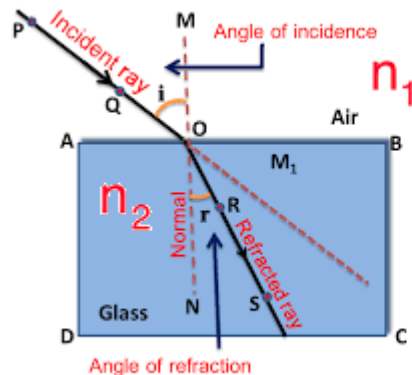


index of refraction

$$n = \frac{c}{v}$$

velocity of light in vacuum

velocity of light in the medium

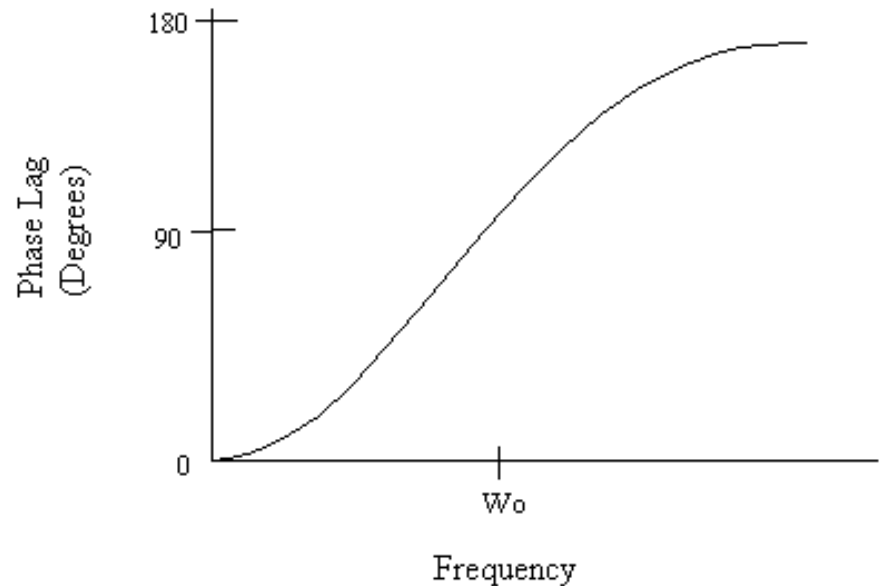
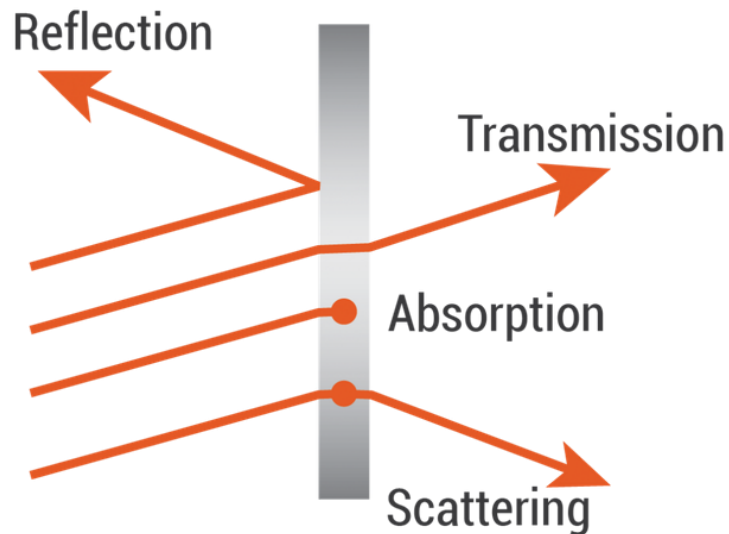


Snell's Law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Transmission:

- When the light is scattered, the wavelets interfere constructively in the forward direction and form the secondary wave front.
- The superposition of the primary and secondary wave front (both traveling at speed c) is the transmitted wave.
- At low frequencies, the molecules are able to vibrate almost in phase with the light. However, as the frequency increases, the vibration of the molecules lag behind, and the phase lag increases with increased frequency



The Blue Sky:

Rayleigh scattering increases with frequency. Therefore, blue light is scattered more than red light. When light from the sun interacts with our atmosphere, the lateral scattering of blue light is much greater than the lateral scattering of the rest of the visible spectrum, and this is partly why the sky appears blue